MATH 15200: Calculus II
Midterm 1

Tuesday, October 27, 2026

Name: _____

ANSWER KEY

Instructions. You have 80 minutes. No notes, books, calculators, phones, or other electronic devices are allowed. Show all of your work and justify your answers; a correct answer without supporting work may not receive full credit. Give exact answers (such as $\frac{\sqrt{2}}{3}$ or 5π), not decimal approximations. The last page is extra space. If you run out of room on a problem, you may continue your answer there; if you do, write on that problem's page that your answer continues on the last page, and label your work on the last page with the problem number.

Problem	**Points**	**Score**
1	20	
2	15	
3	15	
4	10	
5	15	
6	25	
Total	100	

1. (20 points)

(a) (10 points) Prove directly from the ε–δ definition that

$$\lim_{x\to 3}\left(x^2 - 2x\right) = 3.$$

Solution. Let $\varepsilon > 0$ and set $\delta = \min\{1, \varepsilon/5\}$. Suppose $0 < |x-3| < \delta$. Since $|x-3| < 1$, we have $2 < x < 4$, so $3 < x+1 < 5$ and $|x+1| < 5$. Therefore

$$\left|(x^2 - 2x) - 3\right| = |x-3|\,|x+1| \le 5|x-3| < 5\delta \le \varepsilon.$$

(b) (10 points) Define a function f by

$$f(x) = \begin{cases} 3, & x \neq 1, \\ 5, & x = 1. \end{cases}$$

Prove directly from the definition that f is not continuous at 1.

Solution. Since $f(1) = 5$, we must show that $\lim_{x\to 1} f(x) = 5$ is false: that there is an $\varepsilon > 0$ such that for every $\delta > 0$, some x with $0 < |x-1| < \delta$ has $|f(x) - 5| \ge \varepsilon$. Take $\varepsilon = 1$. Given $\delta > 0$, let $x = 1 + \delta/2$. Then $0 < |x-1| < \delta$ and $x \neq 1$, so $|f(x) - 5| = |3 - 5| = 2 \ge \varepsilon$. Hence f is not continuous at 1. (Also valid, if uniqueness of limits was proved in class: $\lim_{x\to 1} f(x) = 3$, since any δ works, and $3 \neq f(1)$.)

2. (15 points) Let $f(x) = 1/x$ on $[1, 2]$, and let $P = \{1, \frac{4}{3}, \frac{5}{3}, 2\}$.

(a) (10 points) Compute the lower sum $L_f(P)$ and the upper sum $U_f(P)$.

Solution. Each subinterval has length $\frac{1}{3}$. Since f is decreasing on $[1, 2]$, its minimum on each subinterval is at the right endpoint and its maximum is at the left endpoint. Hence

$$L_f(P) = \frac{1}{3}\left(\frac{3}{4} + \frac{3}{5} + \frac{1}{2}\right) = \frac{1}{3} \cdot \frac{37}{20} = \frac{37}{60}, \qquad U_f(P) = \frac{1}{3}\left(1 + \frac{3}{4} + \frac{3}{5}\right) = \frac{1}{3} \cdot \frac{47}{20} = \frac{47}{60}.$$

(b) (5 points) Use part (a) to find numbers A and B such that

$$A \leq \ln(2) \leq B,$$

and explain why these inequalities hold.

Solution. Take $A = L_f(P)$ and $B = U_f(P)$. By the definition of the natural logarithm, $\ln 2 = \int_1^2 \frac{dx}{x}$ (or by the Fundamental Theorem of Calculus, since $\frac{d}{dx} \ln x = \frac{1}{x}$), and the integral lies between every lower sum and every upper sum, so

$$\frac{37}{60} \leq \ln 2 = \int_1^2 \frac{dx}{x} \leq \frac{47}{60}.$$

3. (15 points) For $x > 0$, let

$$F(x) = \int_x^{x^2} \sqrt{t^3 + 8}\, dt.$$

(a) (10 points) Find $F'(x)$.

Solution. By the Fundamental Theorem of Calculus and the chain rule (as in Homework 2, Problem 1),

$$F'(x) = \sqrt{(x^2)^3 + 8} \cdot 2x - \sqrt{x^3 + 8} \cdot 1 = 2x\sqrt{x^6 + 8} - \sqrt{x^3 + 8}.$$

(b) (5 points) Find an equation of the tangent line to the graph of F at $x = 1$.

Solution. $F(1) = \int_1^1 \sqrt{t^3 + 8}\, dt = 0$ and $F'(1) = 2\sqrt{9} - \sqrt{9} = 3$, so the tangent line is $y = 3(x - 1)$, that is, $y = 3x - 3$.

4. (10 points) Evaluate the following integrals.

(a) (5 points) $\displaystyle\int \cos x \sqrt{1+\sin x}\, dx$

Solution. Let $u = 1 + \sin x$, so $du = \cos x\, dx$. Then

$$\int \cos x \sqrt{1+\sin x}\, dx = \int \sqrt{u}\, du = \frac{2}{3}u^{3/2} + C = \frac{2}{3}(1+\sin x)^{3/2} + C.$$

(b) (5 points) $\displaystyle\int_1^4 \frac{dx}{\sqrt{x}\left(1+\sqrt{x}\right)^2}$

Solution. Let $u = 1 + \sqrt{x}$, so $du = \dfrac{dx}{2\sqrt{x}}$; when $x = 1$, $u = 2$, and when $x = 4$, $u = 3$. Then

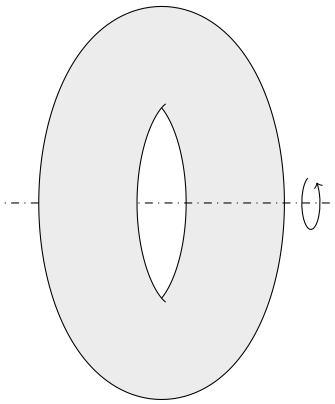
$$\int_1^4 \frac{dx}{\sqrt{x}\left(1+\sqrt{x}\right)^2} = \int_2^3 \frac{2\, du}{u^2} = \left[-\frac{2}{u}\right]_2^3 = -\frac{2}{3} + 1 = \frac{1}{3}.$$

5. (15 points) Find the total (unsigned) area enclosed by the curves $y = x^3$ and $y = x^2 + 2x$.

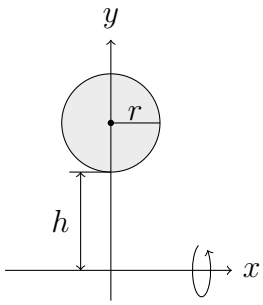
Solution. The curves meet where $x^3 = x^2 + 2x$, that is, $x(x+1)(x-2) = 0$, so at $x = -1$, 0, and 2. On $[-1, 0]$ the cubic is on top (at $x = -\frac{1}{2}$: $-\frac{1}{8} > -\frac{3}{4}$), and on $[0, 2]$ the parabola is on top (at $x = 1$: $3 > 1$). Hence the area is

$$\begin{aligned} A &= \int_{-1}^{0} \left(x^3 - x^2 - 2x\right) dx + \int_{0}^{2} \left(x^2 + 2x - x^3\right) dx \\ &= \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2\right]_{-1}^{0} + \left[\frac{x^3}{3} + x^2 - \frac{x^4}{4}\right]_{0}^{2} = \frac{5}{12} + \frac{8}{3} = \frac{37}{12}. \end{aligned}$$

6. (25 points) Let $r > 0$ and $h > 0$. Revolving the disk $x^2 + \big(y - (h+r)\big)^2 \le r^2$, which has radius r and center $(0, h+r)$, about the x-axis produces a solid called a *torus*; the hole in the middle of the torus has radius h (see the figure). Find the volume of the torus in terms of r and h.



the torus



the disk and the axis

Hint: Use the washer method, and note that $\int_{-r}^{r} \sqrt{r^2 - x^2}\, dx$ is the area of a semicircle of radius r.

Solution. The disk is bounded above and below by the semicircles $y = (h+r) \pm \sqrt{r^2 - x^2}$, for $-r \le x \le r$. So the cross-section of the torus perpendicular to the x-axis at x is a washer with outer radius $(h+r) + \sqrt{r^2 - x^2}$ and inner radius $(h+r) - \sqrt{r^2 - x^2}$, and its area is

$$A(x) = \pi\Big[\big(h + r + \sqrt{r^2 - x^2}\big)^2 - \big(h + r - \sqrt{r^2 - x^2}\big)^2\Big] = 4\pi(h+r)\sqrt{r^2 - x^2}.$$

The integral $\int_{-r}^{r} \sqrt{r^2 - x^2}\, dx$ is the area of a semicircle of radius r, namely $\pi r^2/2$. Hence

$$V = \int_{-r}^{r} 4\pi(h+r)\sqrt{r^2 - x^2}\, dx = 4\pi(h+r) \cdot \frac{\pi r^2}{2} = 2\pi^2 r^2 (h+r).$$

(Shells also work: $V = \int_{h}^{h+2r} 2\pi y \cdot 2\sqrt{r^2 - (y - h - r)^2}\, dy$; after substituting $u = y - (h+r)$, the term $\int_{-r}^{r} u\sqrt{r^2 - u^2}\, du$ vanishes by symmetry.) As a check, $r = 1$ and $h = 2$ give $6\pi^2$, the answer to Homework 3, Problem 2.

Extra Space